

6(a)(i)	line passes through (0,0) and is in first and third quadrants only	M1
	gradient of line becoming less steep in both quadrants and roughly symmetrical	A1
6(a)(ii)	(as I increases) the temperature (of the filament wire/lamp) increases	B1
	(as I / temperature / V increases) the resistance (of wire/lamp) increases	B1
	(as I / temperature / V increases the graph curves because) ratio V/I increases or ratio I/V decreases	B1
6(b)	$R = \rho L / A$	C1
	$= (5.6 \times 10^{-8} \times 5.8) / 3.4 \times 10^{-8}$ $= 9.6 \, \Omega$	A1

Question	Answer					Marks																														
6(c)	<table><tr><td colspan="2">position of switches</td><td colspan="3">ammeter readings</td></tr><tr><td>S₁</td><td>S₂</td><td>X / A</td><td>Y / A</td><td>Z / A</td></tr><tr><td>open</td><td>open</td><td>0</td><td>0</td><td>0</td></tr><tr><td>open</td><td>closed</td><td>1.0</td><td>0</td><td>1.0</td></tr><tr><td>closed</td><td>open</td><td>2.0</td><td>2.0</td><td>0</td></tr><tr><td>closed</td><td>closed</td><td>3.0</td><td>2.0</td><td>1.0</td></tr></table>					position of switches		ammeter readings			S ₁	S ₂	X / A	Y / A	Z / A	open	open	0	0	0	open	closed	1.0	0	1.0	closed	open	2.0	2.0	0	closed	closed	3.0	2.0	1.0	B4
	position of switches		ammeter readings																																	
	S ₁	S ₂	X / A	Y / A	Z / A																															
	open	open	0	0	0																															
	open	closed	1.0	0	1.0																															
	closed	open	2.0	2.0	0																															
	closed	closed	3.0	2.0	1.0																															
second row: both values correct (B1)																																				
third row: all three values correct (B1)																																				
fourth row: $X = Y + Z$ (any values) (B1)																																				
all three values correct (B1)																																				